

The diagram illustrates the architecture of a Kubernetes cluster for e-gov services, organized into three main sections: On-Premise, Kubernetes-Cluster (Behörde), and Cloud (AWS).

- On-Premise, Anbieter durch Behörde gewählt:** This section contains the **F5-Firewall / Reverse Proxy**, which handles TLS termination and sets the **EGOV_TOKEN-Cookie**.
- Kubernetes-Cluster (Behörde):** This section is enclosed in a dashed box and contains:
 - Fachapplikations-Pod:** A blue box representing the application pod, which includes:
 - e-gov-framework (eingebettet):** A light blue box representing the embedded framework, which is JAR-dependent and includes IAM and Command components. It handles **Token / Redirect** and **HTTPS (Login)**.
 - Liveness / Readiness Probe:** A label indicating the pod's health checks.
 - Datenbank-Server:** A yellow box representing the database server, which can be PostgreSQL or SQL Server, containing the framework and domain tables. It connects to the application pod via **JDBC**.
 - ArgoCD:** A purple box representing the GitOps deployment management tool. It handles **Rolling Deploy** and **HTTPS / Maven (beim Start)**.
 - ZooKeeper (optional):** A light gray dashed box representing a distributed service discovery for multi-instance commands. It connects to the application pod via **TCP 2181 (optional)** and receives **Git Push → ArgoCD** notifications.
- Cloud (AWS):** This section contains the **Amazon S3** bucket, which stores **Static Content** and is accessed via **Static Content S3**.

The flow of data and control is as follows:

- The **F5-Firewall / Reverse Proxy** receives external traffic and forwards it to the **Fachapplikations-Pod**.
- The **Fachapplikations-Pod** interacts with the **Datenbank-Server** via **JDBC** and the **ArgoCD** via **Rolling Deploy**.
- The **ArgoCD** manages the deployment of the **Fachapplikations-Pod** and receives **Git Push → ArgoCD** notifications from the **ZooKeeper**.
- The **Fachapplikations-Pod** also interacts with the **Amazon S3** bucket via **Static Content S3**.

